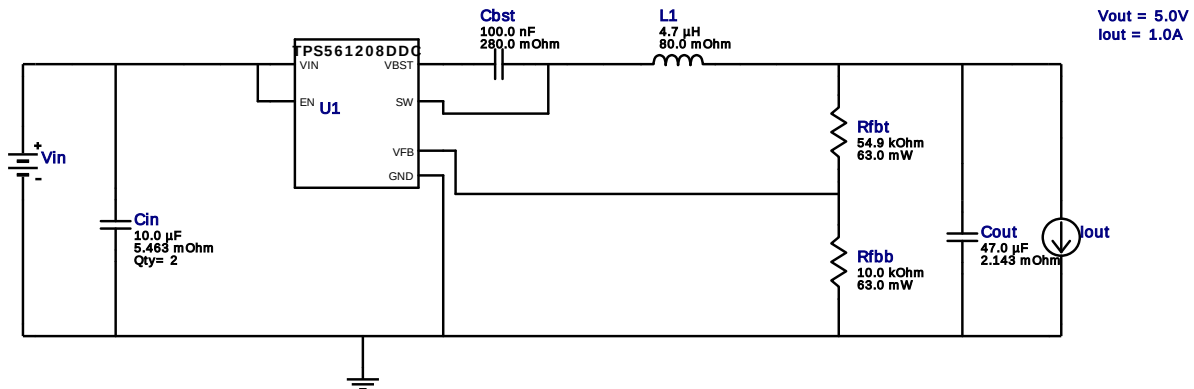


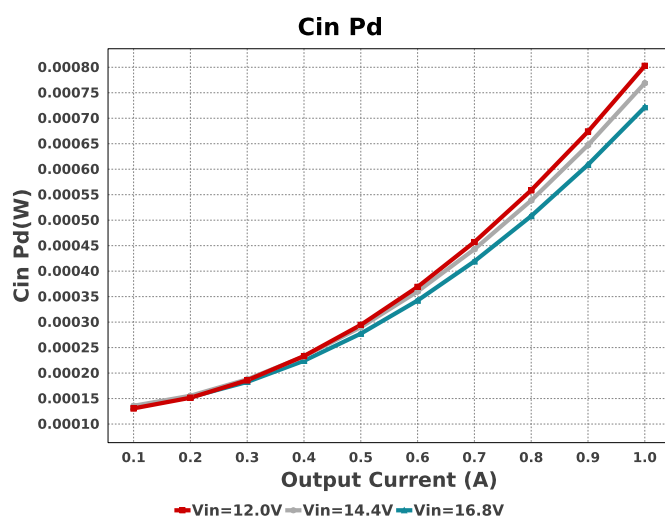
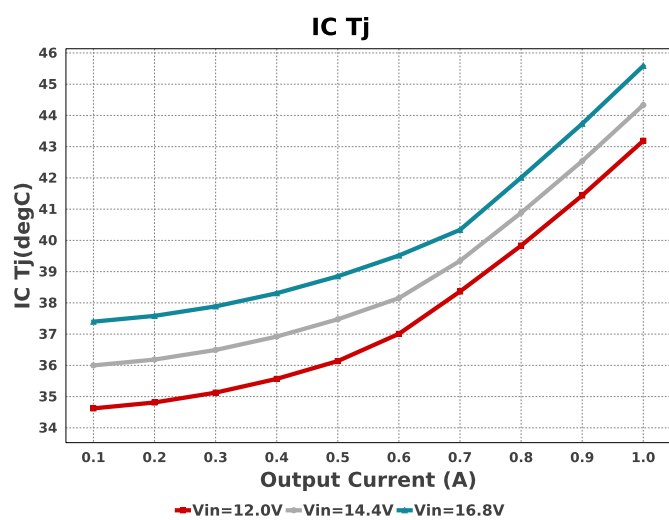
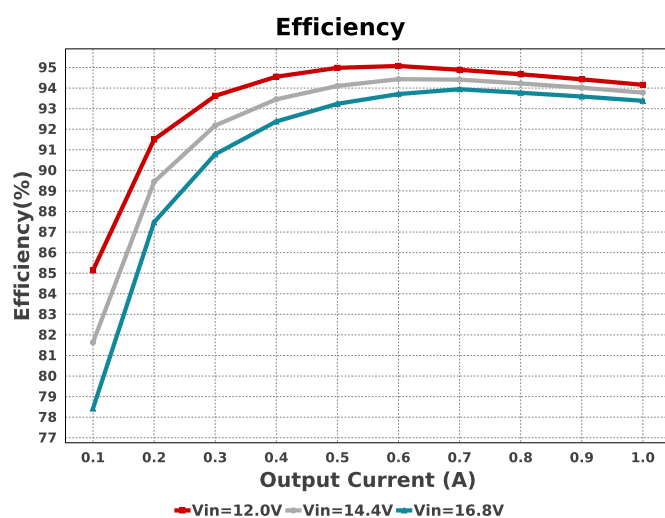
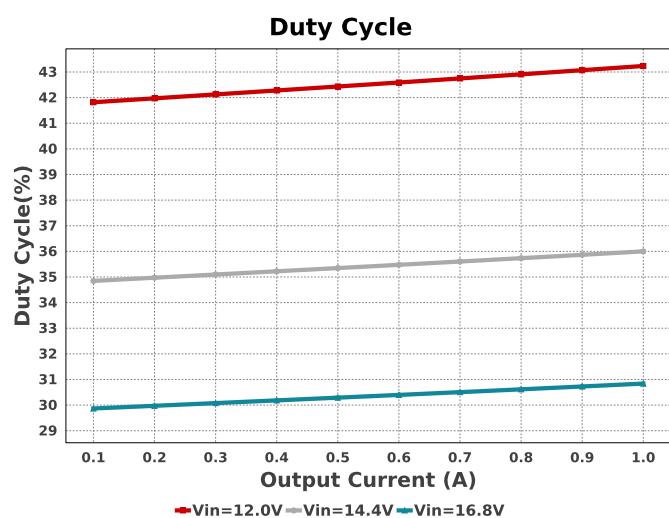
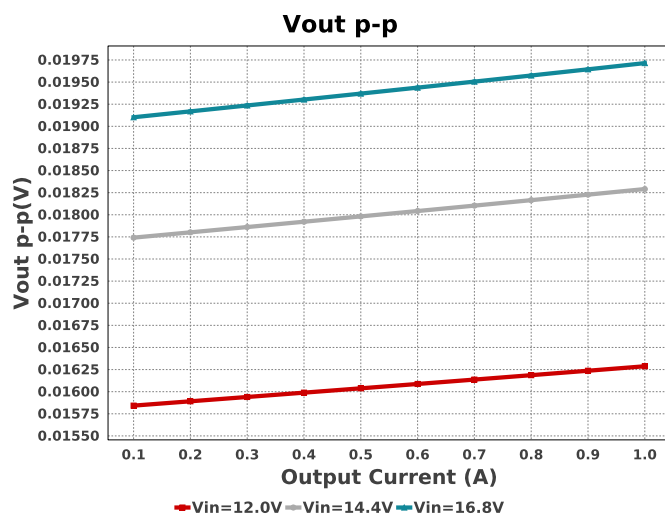
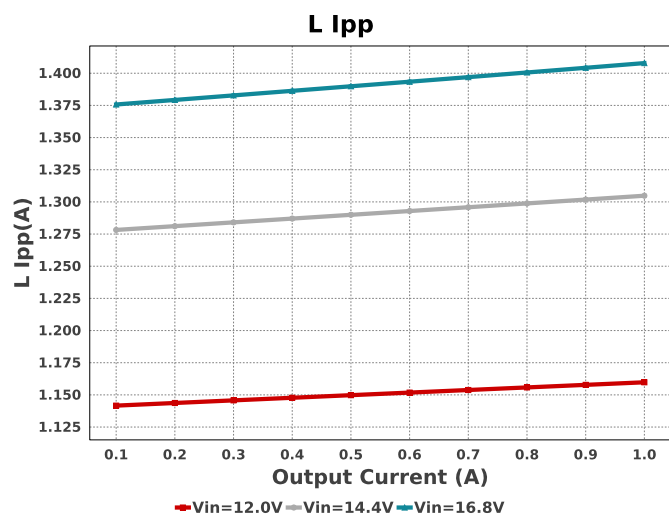
WEBENCH® Design Report

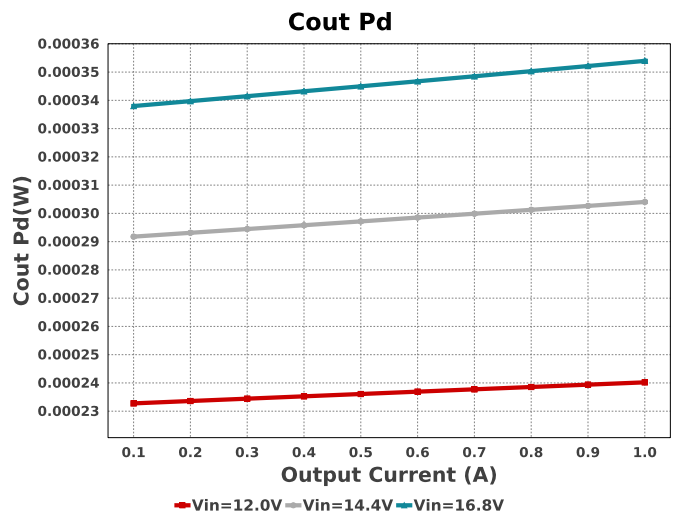
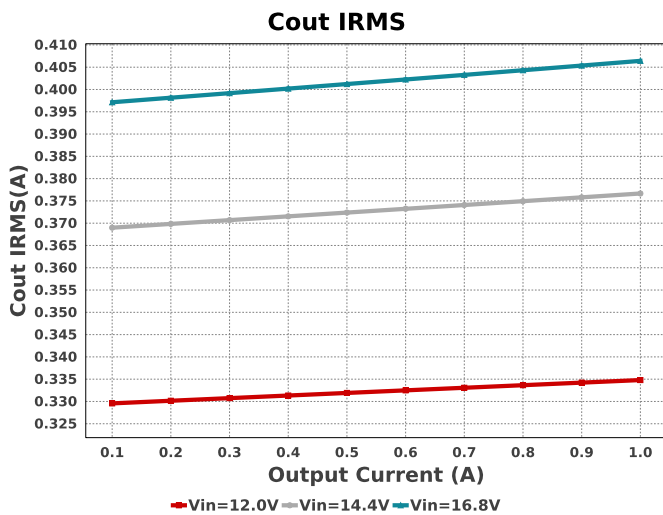
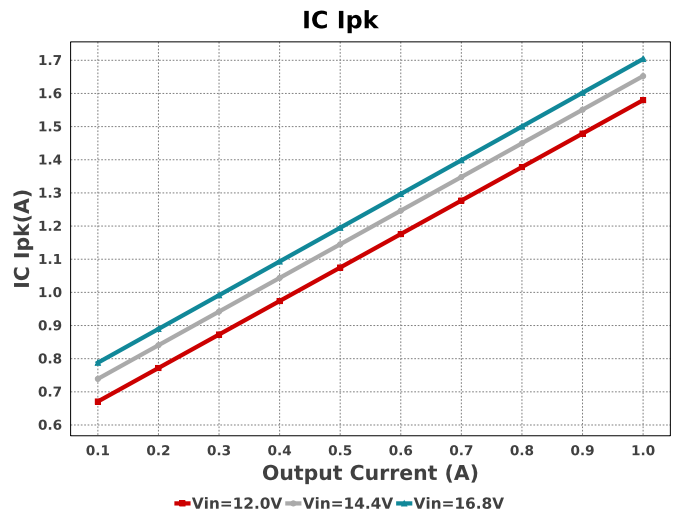
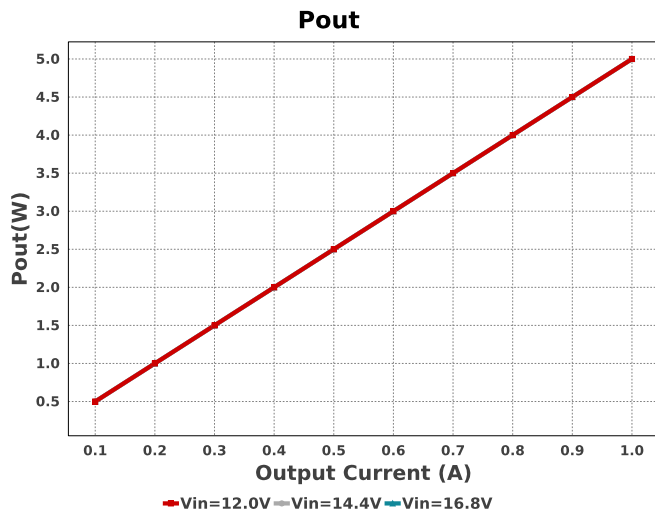
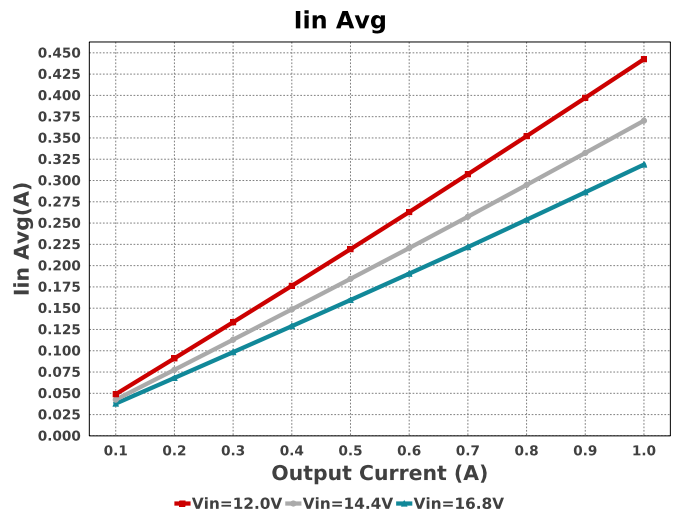
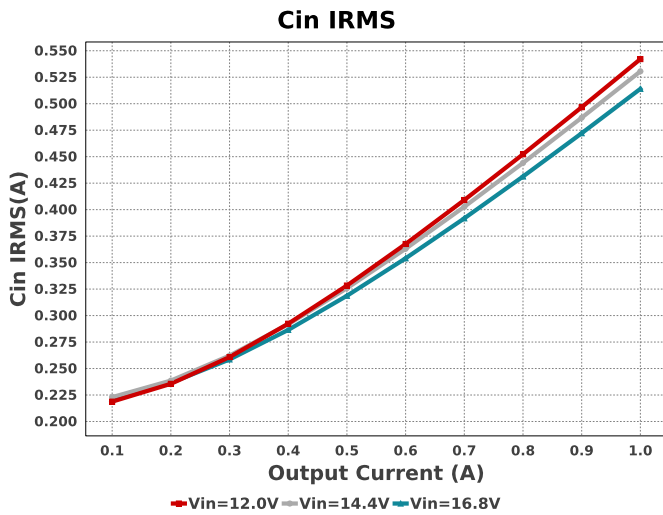
Design : 4 TPS561208DDCR
TPS561208DDCR 12V-16.8V to 5.00V @ 1A

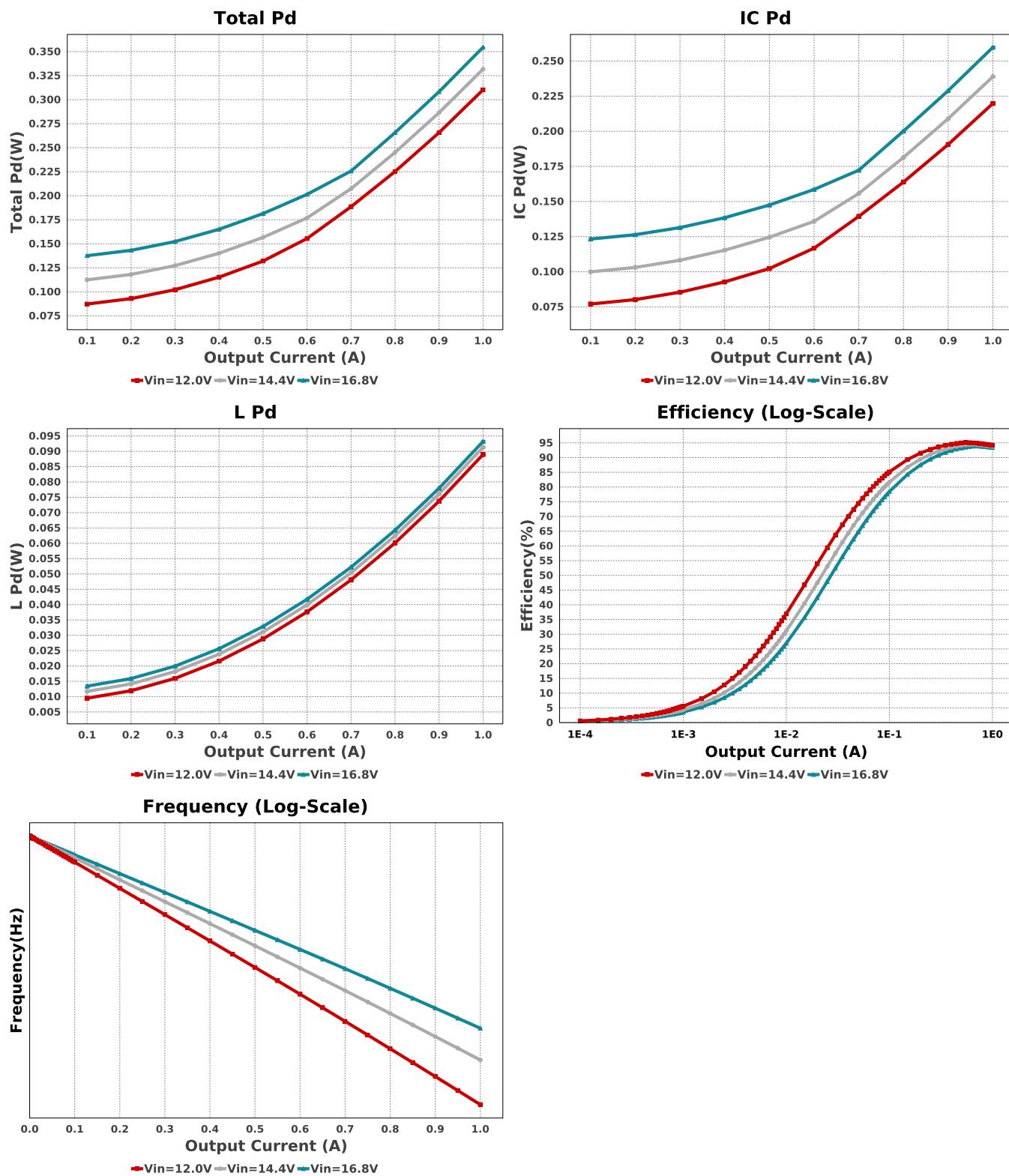


Electrical BOM

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cbst	AVX	08053C104KAT2A Series= X7R	Cap= 100.0 nF ESR= 280.0 mOhm VDC= 25.0 V IRMS= 0.0 A	1	\$0.01	0805 7 mm ²
Cin	Taiyo Yuden	MSAST21GBB5106MTNA01 Series= X5R	Cap= 10.0 uF ESR= 5.463 mOhm VDC= 25.0 V IRMS= 2.36768 A	2	\$0.04	0805 7 mm ²
Cout	TDK	C3216X5R1C476M160AB Series= X5R	Cap= 47.0 uF ESR= 2.143 mOhm VDC= 16.0 V IRMS= 0.0 A	1	\$0.28	1206 11 mm ²
L1	NIC Components	NPI43C4R7MTRF	L= 4.7 uH 80.0 mOhm	1	\$0.14	IND_NPI43C 31 mm ²
Rfbb	Vishay-Dale	CRCW040210K0FKED Series= CRCW..e3	Res= 10.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
Rfht	Vishay-Dale	CRCW040254K9FKED Series= CRCW..e3	Res= 54.9 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
U1	Texas Instruments	TPS561208DDCR	Switcher	1	\$0.20	DDC0006A 15 mm ²







Operating Values

#	Name	Value	Category	Description
1.	Cin IRMS	514.029 mA	Capacitor	Input capacitor RMS ripple current
2.	Cin Pd	721.73 μ W	Capacitor	Input capacitor power dissipation
3.	Cout IRMS	406.42 mA	Capacitor	Output capacitor RMS ripple current
4.	Cout Pd	353.97 μ W	Capacitor	Output capacitor power dissipation
5.	IC Ipk	1.704 A	IC	Peak switch current in IC
6.	IC Pd	259.77 mW	IC	IC power dissipation
7.	IC Tj	45.586 degC	IC	IC junction temperature
8.	IC Tolerance	19.0 mV	IC	IC Feedback Tolerance
9.	ICThetaJA Effective	60.0 degC/W	IC	Effective IC Junction-to-Ambient Thermal Resistance
10.	Iin Avg	318.72 mA	IC	Average input current
11.	L Ipp	1.408 A	Inductor	Peak-to-peak inductor ripple current

#	Name	Value	Category	Description
12.	L Pd	93.214 mW	Inductor	Inductor power dissipation
13.	Cin Pd	721.73 μ W	Power	Input capacitor power dissipation
14.	Cout Pd	353.97 μ W	Power	Output capacitor power dissipation
15.	IC Pd	259.77 mW	Power	IC power dissipation
16.	L Pd	93.214 mW	Power	Inductor power dissipation
17.	Total Pd	354.468 mW	Power	Total Power Dissipation
18.	BOM Count	8	System	Total Design BOM count
			Information	
19.	Duty Cycle	30.839 %	System	Duty cycle
			Information	
20.	Efficiency	93.38 %	System	Steady state efficiency
			Information	
21.	FootPrint	83.0 mm ²	System	Total Foot Print Area of BOM components
			Information	
22.	Frequency	539.342 kHz	System	Switching frequency
			Information	
23.	Iout	1.0 A	System	Iout operating point
			Information	
24.	Mode	CCM	System	Conduction Mode
			Information	
25.	Pout	5.0 W	System	Total output power
			Information	
26.	Total BOM	\$0.72	System	Total BOM Cost
			Information	
27.	Vin	16.8 V	System	Vin operating point
			Information	
28.	Vout	5.0 V	System	Operational Output Voltage
			Information	
29.	Vout Actual	4.984 V	System	Vout Actual calculated based on selected voltage divider resistors
			Information	
30.	Vout Tolerance	4.225 %	System	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
			Information	
31.	Vout p-p	19.715 mV	System	Peak-to-peak output ripple voltage
			Information	

Design Inputs

Name	Value	Description
Iout	1.0	Maximum Output Current
VinMax	16.8	Maximum input voltage
VinMin	12.0	Minimum input voltage
VinTyp	14.8	Typical input voltage
Vout	5.0	Output Voltage
base_pn	TPS561208	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of C_{in} and C_{out} , and the inductance and DC resistance of L1 before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

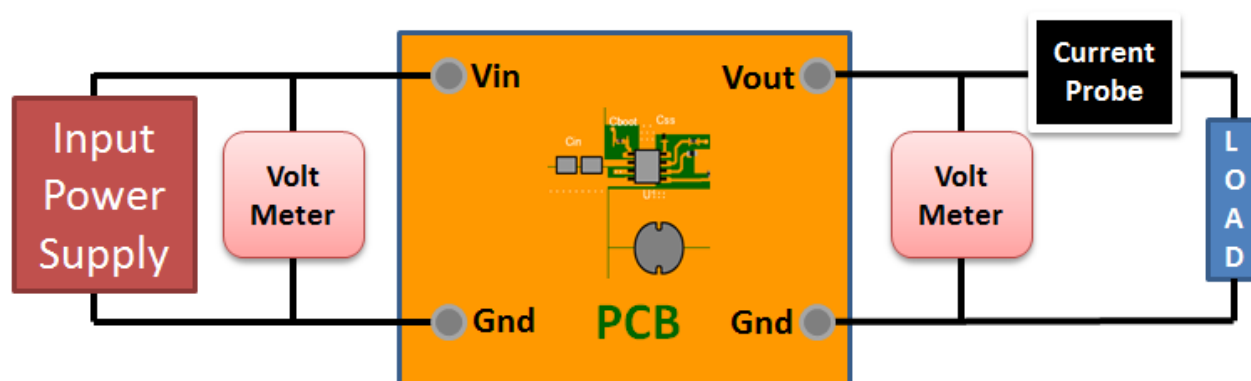
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab down to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 12.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to V_{in} and GND. Connect a digital volt meter and a load if needed to set the minimum load of the design from V_{out} and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between V_{in} and GND, a load is connected between V_{out} and GND and a current meter is connected in series between V_{out} and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. Master key : 84053E93FED573F79AD11CC824948899[v1]
2. **TPS561208** Product Folder : <http://www.ti.com/product/TPS561208> : contains the data sheet and other resources.

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